

Polygon Proof of Stake (0x00)

Network: Polygon PoS Mainnet (Chain ID: 137)

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ELEVATED RISK (UNVERIFIED CODE) (Risk: 72/100 | Quality: 64/100)

STOP-SELL DIRECTIVE: INSUFFICIENT EVIDENCE: Coverage below minimum threshold for institutional sale

20%

[VERIFIED - IMMUTABLE]

Standard: Merkle Non-Repudiation Seal | RFC 3161 Pinned | Verification: </api/audit/report-pdf>

Contract lacks verified on-chain runtime bytecode. In accordance with Velmère evidence standards, missing bytecode results in an elevated risk assessment (72/100).

■ ANALYSIS TYPE: AUTOMATED STATIC & FORMAL ANALYSIS

Rule Governance: Distributed node consensus across validators / miners

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Verification of Polygon Proof of Stake is conducted against native Layer-1 consensus specifications. EVM bytecode decompilation is NOT APPLICABLE for native base protocol assets.

■ **ANALYSIS TYPE: AUTOMATED STATIC & FORMAL ANALYSIS**

Identified risk vectors in base layer architecture

PRO

[LOCKED SECTION - REQUIRES PRO ENTITLEMENT]

- Distributed consensus governance and protocol upgrade mechanism analysis
- Validator / miner distribution and network finality threshold evaluation
- Hard-fork coordination history and client implementation diversity audit
- WHY: Section operates on dedicated compute clusters or requires manual human analyst allocation.
- BLOCKER: TIER ENTITLEMENT BOUNDARY (client active tier does not cover this analytical layer).

Protected evidence and findings withheld pending tier upgrade.

PRO

[LOCKED SECTION - REQUIRES PRO ENTITLEMENT]

- Global order book depth across regulated spot venues and custodial reserves
- On-chain circulating supply velocity and active address distribution
- Network transaction fee volatility and mempool congestion behavior
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PRO

VELMÈRE BASIC AUDIT REPORT

[illegible]

[LOCKED]

[LOCKED SECTION - REQUIRES PRO ENTITLEMENT]

Summary of Analytical Depth in this section:

- Consensus reorganization and majority hashrate/stake attack modeling
- P2P network partition, eclipse attack, and sybil resistance analysis
- Mempool front-running, transaction censorship, and MEV extraction vectors
- WHY: Section operates on dedicated compute clusters or requires manual human analyst allocation.
- BLOCKER: TIER_ENTITLEMENT_BOUNDARY (client active tier does not cover this analytical layer).

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ADVANCED

[LOCKED]

[LOCKED SECTION - REQUIRES ADVANCED ENTITLEMENT]

Summary of Analytical Depth in this section:

- Reference node client reproducible build verification against repository tags
- Consensus serialization and wire protocol binary compatibility analysis
- State transition engine deterministic execution boundary checks
- WHY: Section operates on dedicated compute clusters or requires manual human analyst allocation.
- BLOCKER: TIER_ENTITLEMENT_BOUNDARY (client active tier does not cover this analytical layer).

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ADVANCED

[LOCKED]

[LOCKED SECTION - REQUIRES ADVANCED ENTITLEMENT]

Summary of Analytical Depth in this section:

- Protocol upgrade proposal activation and deployment history
- Backward compatibility verification of block validation logic
- Formal verification: NOT EXECUTED (Formal verification engine not commissioned)
- WHY: Section operates on dedicated compute clusters or requires manual human analyst allocation.
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ADVANCED

[LOCKED]

[LOCKED SECTION - REQUIRES ADVANCED ENTITLEMENT]

Summary of Analytical Depth in this section:

- Analyst sign-off requires verified reviewer evidence
- Cryptographically signed analyst attestation digest
- Manual review boundary explicit: no guarantee of absolute perfection
- WHY: Section operates on dedicated compute clusters or requires manual human analyst allocation.
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